ANANYA SRIDHAR

Speech AI & Audio ML Engineer — ASR, TTS & Acoustic Processing

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GitHub: github.com/darayavahoush · Portfolio: darayavahoush.github.io · LinkedIn: linkedin.com/in/ananya-sridhar-ai · Vaaksudhi Live · Vaakify Live

Languages: English, Hindi, Tamil

PROFESSIONAL SUMMARY

Speech AI and Audio Machine Learning Engineer with hands-on production expertise building low-latency ASR, TTS, and phoneme-level acoustic verification pipelines across 8+ Indian languages. Proven track record deploying clinical speech therapy platforms (Manas Learning) and fintech conversational audio intelligence systems serving 10,000+ financial advisors (Nivesh AI). Skilled in optimizing Whisper, Vosk, and Kokoro ONNX models, audio DSP (Librosa, Praat), FastAPI microservices, and Azure Cloud deployment.

TECHNICAL SKILLS

- **Speech AI & ASR/TTS:** Faster-Whisper, OpenAI Whisper, Vosk, Coqui TTS, Kokoro ONNX, Sarvam AI, Phoneme Verification, Epitran (G2P), Pronunciation Scoring, Forced Alignment, ONNX Runtime
- **Audio DSP & Signal Processing:** Librosa, Praat-Parselmouth, SoundFile, Spectral Noise Reduction (`noisereduce`), Pitch & Formant Tracking, Acoustic Feature Extraction
- **Generative AI & Agentic Systems:** LangGraph, LangChain, RAG (Retrieval-Augmented Generation), ChromaDB Vector Search, Prompt Engineering, PyTorch, Hugging Face Transformers
- **Backend, Cloud & DevOps:** Python, C++, SQL, FastAPI, REST APIs, Microservices, WebSockets, Azure Cloud, Docker, PostgreSQL, GitHub Actions CI/CD, Linux, Uvicorn

PROFESSIONAL EXPERIENCE

AI/ML ENGINEER INTERN (SPEECH AI & FULL-STACK) | MANAS LEARNING

Mar 2026 – Present | Remote / Mumbai, India

- **Speech Platform Architecture:** Architected and deployed full-stack speech therapy platforms (vaaksudhi.manaslearning.com and vaakify.manaslearning.com) delivering clinical assessments across 8 Indian languages.
- **Real-Time Pronunciation Engine:** Built an end-to-end pronunciation scoring pipeline chaining Faster-Whisper, Vosk, and Kokoro/Coqui TTS with phoneme-level acoustic verification (Librosa, Epitran G2P) for instant articulatory feedback.
- **Low-Latency Audio Streaming:** Optimized model inference pipelines using ONNX Runtime and streaming audio buffers, achieving sub-300ms round-trip latency over WebSockets.
- **Agentic Diagnostic Workflows:** Implemented LangGraph multi-agent pipelines coordinating 4 diagnostic assessment modules, evaluating place/manner/voicing speech errors and auto-generating clinician reports.
- **Adaptive Reinforcement Learning:** Formulated a Q-learning difficulty controller dynamically tuning therapy exercises in real time based on rolling acoustic accuracy and response latency.
- **Production APIs & Security:** Engineered FastAPI REST APIs and PostgreSQL schemas; integrated COPPA-compliant parental consent flows and secure JWT session authentication.

AI/ML ENGINEER INTERN (AUDIO ML & RAG) | NIVESH AI

Jun 2026 – Present | Mumbai, India

- **Fintech Speech Ingestion Infrastructure:** Engineered scalable audio ingestion and processing microservices on Azure Cloud, supporting wealth-management intelligence for 10,000+ mutual fund distributors (MFDs).
- **Multilingual Speech-to-Text Optimization:** Integrated and fine-tuned domain-adapted STT models (Whisper, Sarvam AI) with spectral denoising (`noisereduce`), boosting transcription fidelity on mixed Indian English and Hindi audio.
- **Zero-Hallucination RAG Pipeline:** Developed an enterprise RAG summarization engine using ChromaDB vector search and attention-based LLMs to extract actionable portfolio insights from advisor calls.
- **Automated Workflow Impact:** Slashed advisor documentation turnaround time by 80%, automating manual note-taking and compliance tracking across tens of thousands of client interactions.
- **Cloud Reliability & DevOps:** Containerized backend services with Docker on Azure, set up asynchronous task workers, and implemented automated API health checks and structured logging.

FEATURED PROJECTS

VAAKSUDHI — PEDIATRIC SPEECH & DIAGNOSTIC MULTI-AGENT PLATFORM

Live Platform · GitHub

Python, LangGraph, Faster-Whisper, Vosk, Kokoro ONNX, Librosa, Epitran, Q-Learning, FastAPI, React

- Architected an autonomous multi-agent clinical diagnostic platform evaluating speech errors across 8 Indian languages with sub-300ms phoneme alignment.
- Implemented LangGraph state machines and acoustic biofeedback algorithms to auto-generate objective clinical reports for speech therapists.
- Live in active clinical use with automated error taxonomy categorization and personalized practice games.

VAAKIFY — PEDIATRIC MULTILINGUAL SPEECH PRACTICE PLATFORM

Live Platform · GitHub

Python, Librosa, Praat-Parselmouth, Faster-Whisper, Coqui TTS, FastAPI, React

- Built an interactive speech practice platform featuring animated companions and real-time phoneme biofeedback for children with speech challenges.

- Extracted acoustic biomarkers (formants, jitter, shimmer, pitch contours) to identify articulatory drift and dynamically construct personalized corrective drills in English, Hindi, and Kannada.
- Deployed clinical companion actively used by children and therapists for gamified phoneme therapy with objective biofeedback.

NYAYAI & AI BAIL RECKONER — LEGAL-REASONING LLM & RAG ASSISTANT

GitHub Repository

PyTorch, Hugging Face Transformers, ChromaDB, FastAPI, React

- Developed a domain-tuned legal AI assistant predicting bail hearing outcomes and citing judicial precedents across IPC and BNS statutes.
- Implemented a dense RAG pipeline using ChromaDB vector embeddings to ground LLM reasoning strictly in verified Supreme Court rulings, eliminating hallucinations.
- Accelerated legal precedent retrieval from hours to seconds with verifiable case citations.

TRIPWIRE — EDGE COMPUTER VISION VEHICLE COUNTER

Live Demo · GitHub

Python, OpenCV, FastAPI, React, Vite, WebSockets, Docker

- Engineered an edge-ready, CPU-only vehicle detection and tracking pipeline from live CCTV feeds with sub-50ms latency.
- Implemented Gaussian background subtraction, centroid tracking, and directional virtual tripwire algorithms on commodity CPUs without requiring GPUs.
- Shipped a low-latency monitoring dashboard streaming processed video over MJPEG and traffic crossing counts via WebSockets.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering

SASTRA Deemed University, India

Expected Graduation: 2027

CERTIFICATIONS & RECOGNITIONS

- **SWAYAM – IIT Madras:** Comprehensive Electronics and Embedded Systems *(In-Person Practical Program)*
- **IIoT Certification:** Ministry of Heavy Industries (MHI) Program *(Project-Based Evaluation)*